

6.5

Average Value of a Function



In this section, we will learn about:
Applying integration to find out
the average value of a function.

It is easy to calculate the average value of finitely many numbers

$y_1, y_2, \ldots, y_n$:

$$y_{ave} = \frac{y_1 + y_2 + \cdots + y_n}{n}$$

However, how do we compute the average temperature during a day if infinitely many temperature readings are possible?



So, we define the average value of f on the interval $[a, b]$ as:

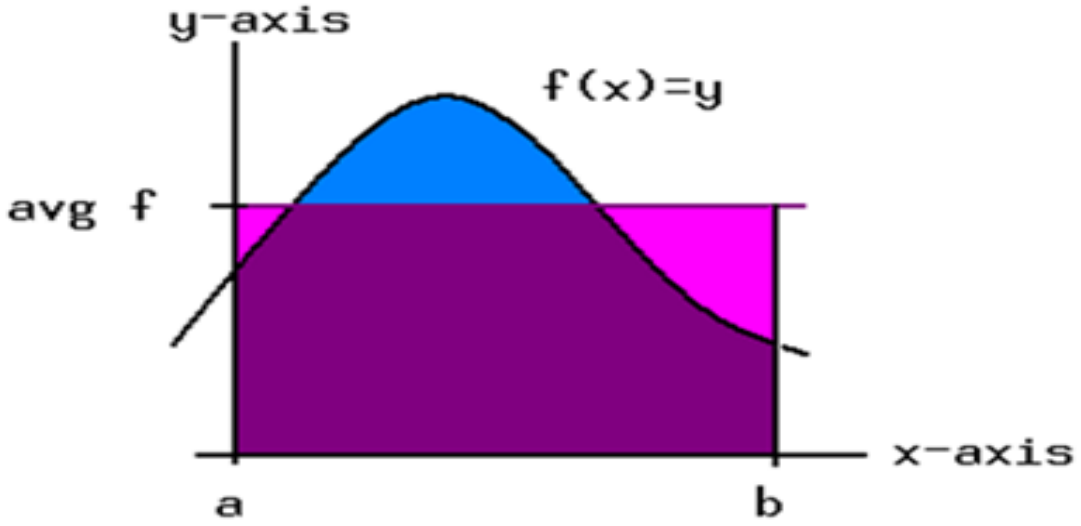
$$f_{ave} = \frac{1}{b-a}\int_a^b f(x)dx$$



Example:

- **Find the average value of the function $f(x) = 1 + x^2$ on the interval [-1, 2].**







In general, is there a number *c* at which the value of a function *f* is exactly equal to the average value of the function—that is, $f(c) = f_{ave}$?



MEAN VALUE THEOREM

If f is continuous on $[a, b]$, then there exists a number c in $[a, b]$ such that

$$f(c) = f_{ave} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

that is,

$$\int_a^b f(x)\,dx = f(c)(b-a)$$

The geometric interpretation of the Mean Value Theorem for Integrals is as follows.

- **The geometric interpretation of the Mean Value Theorem for Integrals is as follows.**
  - **For ‘positive’ functions *f*, there is a number *c* such that the rectangle with base [*a*, *b*] and height *f*(*c*) has the same area as the region under the graph of *f* from *a* to *b*.**







Find the average value of $f(x)=x^2+x+1 \quad on [1,3]$

Example

Find the average value of

- $f(x)=\cos^4 x \sin x \ \ on [0,\pi]$



- **Find c such that the average value of $f(x)=3x^2-2x \ on [1,4]$**

